

SOFTWARE, NETWORK AND DATA ARCHITECTURAL GUIDELINES

In an effort to drive consistency and a high level of architectural quality in our programs, the portfolio architecture office has established the following guidelines:

- **Prospective and existing technology selections should be listed as Preferred or Acceptable in the UHG tech PADU.**
 - UHG Tech PADU can be found here: <https://techlandscape.uhg.com>
 - Instructions for adding a technology to the tech PADU can be found here: <https://techpadu.uhg.com/KC/TLUpdatingTechPadu/> -- any proposed additions should be discussed with the portfolio architects prior to submission.
 - Existing technology that is listed as discouraged or unacceptable should have a documented plan for replacement and should be reviewed with the portfolio and OI security architects.
 - Preference should be given to inner-source & open-source over licensed 3rd party tools.
- **Enterprise and/or OI platform capabilities should be leveraged when viable.**
 - Enterprise identity providers (OHID, MSID, HSID) is strongly preferred.
- **All solutions should implement zero trust principles.**
 - Azure Zero Trust Guidance: [Zero Trust security in Azure | Microsoft Learn](#)
 - AWS Zero Trust Guidance: [Understanding Zero Trust principles - AWS Prescriptive Guidance \(amazon.com\)](#)
- **Solution architecture should closely follow the guidance of all classic pillars (reliability, security, cost optimization, operational excellence & performance efficiency) within the well-architected framework.**
 - <https://dojo360.optum.com/docs/cloud/azure/architecture/concepts/well-designed-application.html>
 - Azure specific guidance - <https://learn.microsoft.com/en-us/azure/well-architected/>
 - AWS specific guidance - <https://aws.amazon.com/architecture/well-architected>
- **Applications must comply with the security standards set by EIS and described within the eGRC site.**
 - <https://uhg-egrc.archer.rsa.com/apps/ArcherApp/Home.aspx>
- **APIs should follow the guidance outlined in the API Maturity standards.**
 - [Provider API Strategy.pdf](#)
- **Each solution should create and maintain a Solution Intent and review with the Architecture Review Board quarterly.**
 - Solution Intent Template can be found here: [05 Architecture Review Board \(ARB\)](#)
- **When integrating with products or services that cross cloud provider boundaries follow the guidance found here: [Multi Cloud Integration Guidance.pptx](#)**
- **The OI managed Splunk Observability tools should be utilized for logging and monitoring.**
 - <https://dojo360.optum.com/docs/devops/lob/uhg/optum/optuminsight/observability/index.html>